Neural Collapse

Vlad Kobzar[1]

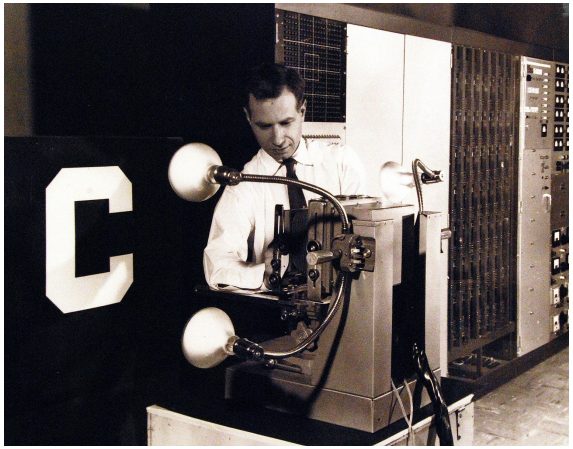
[1] Ohio State

ML paper club

Outline

- Background and motivation
  - Neural Collapse
  - Unconstrained Features Model
- Open questions

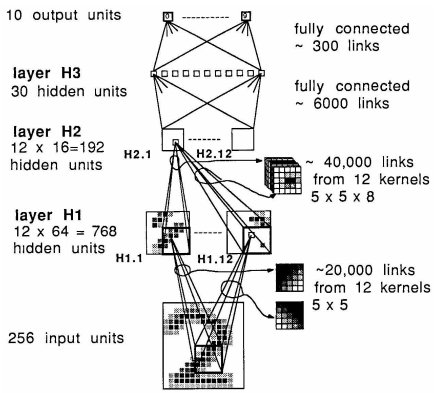
Classification



Mark I Perceptron Announced By Navy (1960)

- Learn a *classifier* $N : \mathcal{X} \to [n]$
  - Space of inputs $\mathcal{X} \subseteq \mathbb{R}^r$
  - Set of labels $[n] := \{1, \ldots, n\}$
  - Training data $\{x_i, k_i\}_{i \in [m]}$ in $\mathcal{X} \times [n]$
- Rich history
  - Handwritten zip code recognition [LeCun et al., 1989]
  - ImageNet [Krizhevsky et al., 2012]
  - DNNs are SOTA in vision & many other problems

Features



Feature maps in LeCun et al. [1989]

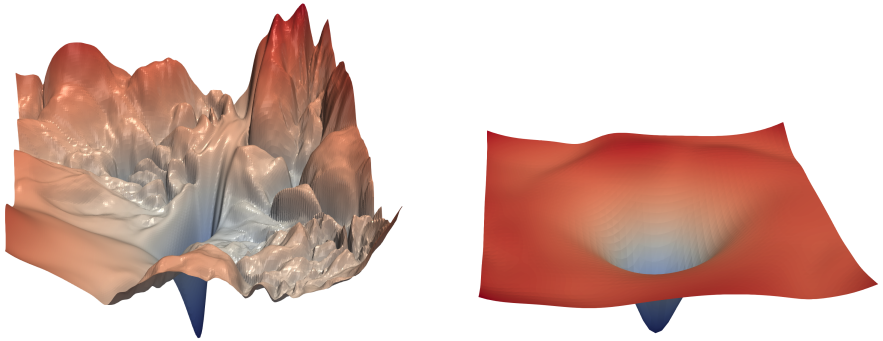
- Typical classifiers $\arg\max_k (L \circ F(x))_k$ are a *feature map*

$$F : \mathcal{X} \to \mathbb{R}^d$$

followed by an affine map $L : \mathbb{R}^d \to \mathbb{R}^n$

- In DL, the feature maps are given by NNs F_θ with weights θ
- Learn θ from the training data
- We'll omit the dependence of F on θ

Empirical risk minimization



Examples of loss surfaces of ResNet (without vs with skip connections) from Li et al. [2018]

- We fit (θ, L) to the training data by minimizing the cross-entropy

$$\mathcal{L}(\theta, L) = -\sum_{i\in[m]}\sum_{j\in[n]}(y_i)_j \log(\text{softmax} \circ L \circ F_\theta(x_i))_j$$

where $y_i := e_{k_i}$ is a *one-hot* encoding of k_i by the canonical basis

Neural collapse [Papyan et al., 2020]

- Train beyond zero misclassification rate: for all $j \in [m]$

$$\arg\max_k (L \circ F(x_i))_k = k_j$$

- For each x_i and $x_{i'}$ with the same label k_j, the features collapse

$$F(x_i) = F(x_{i'}) = h_j$$

- If we have class balance: for each $j \in [n]$,

$$|\{i \in [m] : k_i = j\}| = N$$

  then $\{h_j\}_{j \in [n]}$ are regular simplex in $\mathbb{R}^d$ (pairwise distances equal).
- Assumes these n such points can be realized as such simplex, which requires $d \geq n - 1$.

Alignment of L with F [Papyan et al., 2020]

- The kth entry of the affine linear map $L\colon \mathbb{R}^d \to \mathbb{R}^n$ is given by

$$L(h)_k = \langle w_k, h \rangle + b_k,$$

- In the neural collapse, the weights $\{w_k\}_{k\in[n]}$ in $\mathbb{R}^d$ also form a regular simplex centered at the origin such that

$$w_k \propto h_k - \frac{1}{n}\sum_{i=1}^{n} h_i.$$

Unconstrained features model of Mixon et al. [2022]

- Consider each $h_i \in \mathbb{R}^d$ as a free optimization variable
- Justified since F is generally overparametrized
- Decouples the analysis of NC from the architecture F
- Emergence of geometric features when $d + 2 \leq n \leq 2d$ (orthoplex regime).
- Convexity/concavity of the training loss depends on the regularization parameter (temperature)

Softmax code in the orthoplex regime

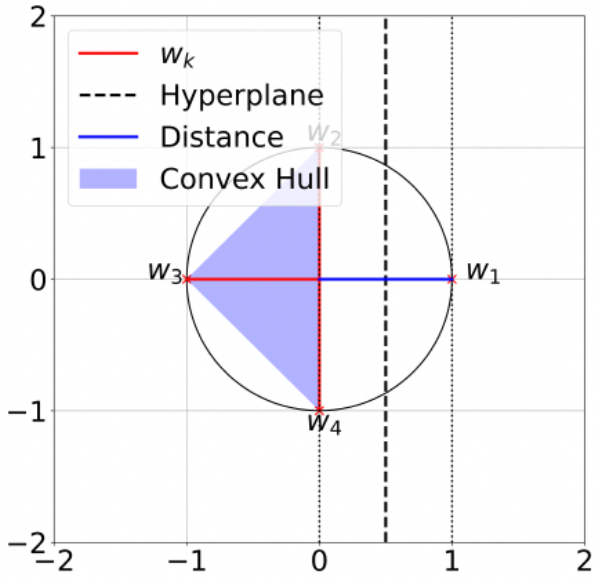


Figure: softmax code for $d = 2, n = 4$

- A softmax code is a configuration of vectors that maximizes the minimal distance between one point and the convex hull of others.

Interpretation of softmax code

- From the structure of softmax code, we can conclude that the learned features are maximally separated, measured by one-vs-rest distance.
- Using the convergence to softmax code, one can benefit in practical problems such as reducing training cost.

Unconstrained features model of Mixon et al. [2022]

- Open questions when $n > 2d$, which is relevant in practical cases
  1. person identification (each identity = one class)
  2. language models (number of classes = vocabulary size)

References I

Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. In F. Pereira, C.J. Burges, L. Bottou, and K.Q. Weinberger, editors, *Advances in Neural Information Processing Systems*, volume 25. Curran Associates, Inc., 2012. URL https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf.

Yann LeCun, Bernhard Boser, John S Denker, Donnie Henderson, Richard E Howard, Wayne Hubbard, and Lawrence D Jackel. Backpropagation applied to handwritten zip code recognition. *Neural computation*, 1:4:541–551, Dec 1989.

References II

Hao Li, Zheng Xu, Gavin Taylor, Christoph Studer, and Tom Goldstein. Visualizing the loss landscape of neural nets. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/file/a41b3bb3e6b050b6c9067c67f663b915-Paper.pdf.

Dustin G. Mixon, Hans Parshall, and Jianzong Pi. Neural collapse with unconstrained features. *Sampling Theory, Signal Processing, and Data Analysis*, 20:2, Jul 2022.

V. Papyan, X. Y. Han, and D. L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proc. Natl. Acad. Sci. U.S.A.*, 117:24652–24663, 2020.